

FRAUDMESH

Original Problem Statement

Project Type: AI-powered financial fraud investigation web platform

Core idea: FraudMesh reconstructs fragmented financial-fraud evidence into an explainable network so investigators can understand how apparently separate fraud incidents may be connected.

1. Problem Statement

Financial fraud is increasingly difficult to investigate because evidence is fragmented across different victims, bank accounts, UPI IDs, phone numbers, devices, transactions, complaints, timestamps and related incidents. Individual systems may flag suspicious transactions or identifiers, but investigators still face the harder task of reconstructing the complete fraud network from scattered evidence.

Two complaints may look unrelated because they involve different victims, amounts, dates or accounts. However, they may share a phone number, UPI ID, device, receiving account, intermediary account, transaction pattern or other entity. Without connecting these relationships, investigators can miss coordinated fraud operations.

2. Core Gap

The central gap is not simply detecting a suspicious transaction. It is **understanding the relationships between fragmented pieces of evidence across cases** and reconstructing the larger fraud network.

Traditional investigation may answer: *“Is this transaction suspicious?”*

FraudMesh aims to answer: *“Are these incidents connected, how are they connected, what evidence supports the connection, and what should investigators examine next?”*

3. Proposed Solution — FraudMesh

FraudMesh is a working web application for authorized investigators or fraud analysts. It combines graph-based analysis, transaction relationships, chronological reconstruction, risk scoring and Nemotron-powered reasoning to turn fragmented evidence into an understandable investigation view.

Core pipeline

Step	Stage	Purpose
1	Evidence Ingestion	Complaints, transactions and other authorized evidence
2	Entity & Link Extraction	Identify accounts, phones, devices, UPI IDs and relationships
3	Fraud Graph	Connect entities and cases into a network
4	Chronology	Reconstruct the sequence of relevant events
5	Risk Scoring	Prioritize suspicious clusters and cases

Step	Stage	Purpose
6	Nemotron Reasoning	Analyze fragmented evidence and explain relationships
7	Investigation Plan	Suggest evidence-backed next steps for human review

4. Key Features

Fraud Network Visualization: Interactive graph showing connections between victims, accounts, UPI IDs, phones, devices, transactions, complaints and cases.

Risk Scoring: Evidence-backed scoring based on relationships, transaction behavior, repeated entities, timing and network characteristics.

Timeline Reconstruction: Chronological view of transactions, complaints and related events.

Explainable AI: AI conclusions should identify the evidence and relationships supporting the finding.

Evidence Explorer: Investigators can inspect the source information behind important findings.

Investigation Planning: Generate prioritized, human-reviewable investigation steps.

Real-time-style updates: New authorized evidence can update relationships, risk and alerts within the application architecture without claiming direct access to live banking infrastructure.

5. Role of Nemotron

Nemotron should provide meaningful reasoning rather than functioning as a generic chatbot. Deterministic software handles exact matching, numeric calculations, database operations, graph construction, filtering and other precision tasks. Nemotron handles difficult reasoning over fragmented or unstructured evidence.

Examples of Nemotron tasks:

- Understanding unstructured complaint text
- Extracting contextual entities and relationships
- Reconciling fragmented evidence
- Detecting contradictions
- Reconstructing chronology
- Explaining why multiple cases may be related
- Producing evidence-grounded investigative hypotheses
- Summarizing complex fraud networks
- Suggesting investigation priorities

6. Example Fraud Chain

A victim sends money to a UPI identifier. The receiving account transfers the funds to another account shortly afterward. A second victim reports a different transaction involving the same intermediate account. The intermediate account and another account share a device identifier, while a phone number connects the UPI identifier to the intermediate account. Individually these records may look separate; together they form a potentially connected network requiring investigation.

7. Output of FraudMesh

FraudMesh should produce:

- Connected fraud network
- Risk score and contributing factors
- Chronological timeline
- Evidence-backed findings
- Related cases/entities
- Explainable AI analysis
- Investigation priorities
- Suggested investigation plan
- Audit trail of investigator activity

8. Human-in-the-Loop

FraudMesh does not autonomously declare people guilty or take irreversible financial or law-enforcement actions. AI output is an investigation indicator, potential relationship, investigative hypothesis, evidence-backed finding or suggested next step. A human investigator remains responsible for final decisions.

9. Data & Privacy

The hackathon implementation should use synthetic or explicitly authorized data. Real personal financial information should not be exposed in the demonstration. Authentication, authorization, secure secret handling, input validation and audit logging should be treated as core requirements.

10. Product Boundary

FraudMesh is a web application/site. It is not a bank, UPI application, payment gateway or payment-blocking service. The project must not claim that the website can directly stop Google Pay, PhonePe, Paytm or other third-party payment transactions. Such capabilities would require an authorized integration with the relevant financial infrastructure.

11. Core Differentiator

FraudMesh is not simply an AI chatbot for fraud detection.

Core differentiator:

Fragmented Evidence + Graph Relationships + Temporal Reconstruction + Risk Scoring + Nemotron Reasoning + Explainable Investigation

12. Final Problem-to-Solution Statement

Problem: Financial-fraud evidence is fragmented across cases, entities, transactions and records, making it difficult to identify coordinated fraud networks and understand the sequence of events.

Solution: FraudMesh connects these fragments into an interactive fraud network, reconstructs timelines, calculates explainable risk, uses Nemotron to reason over complex evidence, and provides

investigators with prioritized, evidence-grounded investigation guidance.

Final Product Definition

FraudMesh is a working AI-powered web platform for financial-fraud investigation that transforms fragmented fraud evidence into connected networks, chronological event views, explainable risk scores and evidence-grounded investigation plans using graph analysis and Nemotron reasoning.